GMI German-Malaysian Institute

DIPLOMA PROGRAMME

LAB WORK 3

Academic Period: January 2026

Program	:	**CYBER SECURITY TECHNOLOGY (CBS)**
Course	:	**PENETRATION TESTING**
Code	:	**CBS 2353**
Total Marks	:	**20 MARKS**

Title	:	**SCANNING** [Enumeration with Nmap]

Student's Name	:	Abdul Hariz bin Mohd Sarnudin (CBS24070208)
TTO's Name	:	Ts. Noormelah Binti Shamsul Anuar
Semester/Trade/Group	:	Semester 4 / DCBS / CBS 1
Date	:	1/4/2026

Assessment Specification Table (CBS)

Lab work

Raw Marks:	100
Percentage (in Doc 01):	30

No	Assessment Criteria	Full Marks	Score (if applicable)	Student's Marks	CLO	Level of Difficulty	DK, DP, NA (ETAC Programme Only)	Comments	PLO GMI: PLO CBS 9 PLO - Start Intake July 2023	PLO GMI: PLO MBOT 9 PLO
1	Understanding of Lab Objectives	4			2	3			2	2
2	Demonstration of Labwork Execution	4			2	3			2	2
3	Accuracy of Findings / Results	4			2	3			2	2
4	Fluency and Clarity of Explanation	4			2	3			2	2
5	Question & Answer (Q&A)	4			2	3			2	2
									#N/A	#N/A
									#N/A	#N/A
									#N/A	#N/A
									#N/A	#N/A
	Total Full Marks	**20**	**Total Student's Marks**							

SUMMARY

Note:
FM - Full Marks,
StM - Student's Mark

	CLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI: PLO CBS 9 PLO - Start Intake July 2023 PLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM	PLO GMI: PLO MBOT 9 PLO	Raw Marks FM	Raw Marks StM	Percentage FM	Percentage StM
Total	CLO 1	0		0.0		PLO 1	0		0.0		PLO 1	0		0.0	
	CLO 2	20		6.0		PLO 2	20		6.0		PLO 2	20		6.0	
	CLO 3	0		0.0		PLO 8	0		0.0		PLO 8	0		0.0	
	CLO 4	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	CLO 5	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	CLO 6	0		0.0		#N/A	0		0.0		#N/A	0		0.0	
	Total Marks	**20**		**6.0**											

Signature

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NO.	CRITERIA	EXCELLENT [4]	GOOD [3]	SATISFACTORY [2]	WEAK/ NOT MET [1]	MARKS
INDIVIDUAL COMPONENT (TOTAL MARK = 20)						
1	**Understanding of Lab Objectives**	Clearly explains the purpose, scope, and objectives of the lab work	Explains objectives with minor gaps	Basic understanding of objectives	Objectives misunderstood or not explained	
2	**Demonstration of Labwork Execution**	Demonstrates lab tasks correctly, confidently, and independently	Demonstration mostly correct with minor errors	Demonstration partially correct or requires guidance	Unable to demonstrate lab tasks correctly	
3	**Accuracy of Findings / Results**	Findings are accurate, relevant, and clearly linked to the task	Mostly accurate findings with minor errors	Limited or partially correct findings	Unable to demonstrate lab tasks correctly	
4	**Fluency and Clarity of Explanation**	Explanation is clear, fluent, well-structured, and uses correct technical terms	Generally clear with minor language or structure issues	Explanation is basic and sometimes unclear	Poor, unclear, or disorganized explanation	
5	**Question & Answer (Q&A)**	Answers all questions confidently and accurately	Answers most questions correctly	Answers some questions with hesitation some questions with hesitation	Unable to answer questions correctly	
				TOTAL	20	

LABWORK 3 – SCANNING

Labwork Requirements

1. Students must follow all procedures stated in the given lab sheet. All tasks should be completed according to the instructions provided.
2. Students are required to take clear screenshots of relevant outputs as proof of task completion. All screenshots must be properly labeled.
3. Students are required to attend a demonstration session. During the session, students must:
    - Briefly explain the labwork,
    - Perform a live demonstration, and
    - Answer questions related to the labwork.

Non-compliance with these requirements may affect the overall marks.

Submission:
Report (pdf)

Grading:

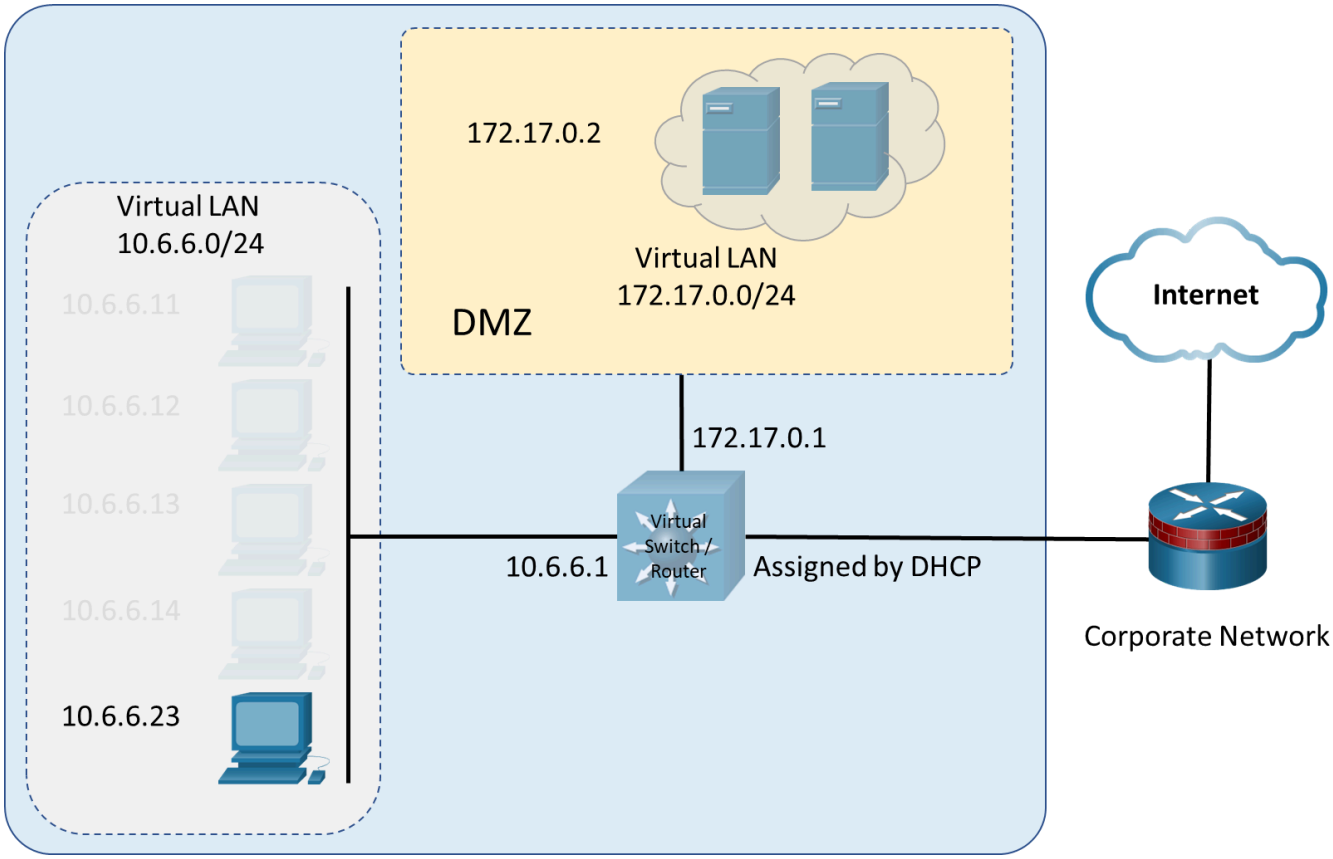
1) Understanding of Lab Objectives
2) Demonstration
3) Accuracy of Findings / Results
4) Fluency and Clarity of Explanation
5) Question & Answer (Q&A)

Team:
Individual

YOU REPORT START ON THE NEXT PAGE.

Labwork #3 – Enumeration with NMAP

Topology



Objectives

Nmap is a powerful open-source tool for network mapping and discovery. In this lab, you will use Nmap as part of your active reconnaissance strategy.

Investigate Nmap

Perform Basic Nmap Scans

Background / Scenario

A Wireshark capture shows unusual activity on a machine on the 10.6.6.0 DMZ network. You've been asked to do some active recon on the machine to determine what services it may be offering and if there are vulnerable applications that could present security issues. The IP address of the suspicious computer is 10.6.6.23. You have access to a Kali Linux system on the 10.6.6.0 network.

Required Resources

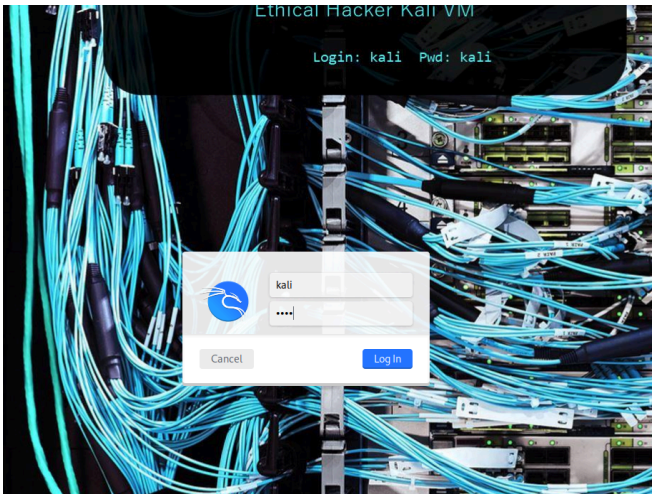
Kali VM customized for Ethical Hacker course

Instructions

Part 1: Investigate Nmap

Step 1:Log into Kali Linux and verify the environment.

a. Log into the Kali system with the username **kali** and the password **kali**. You are presented with the Kali desktop.



b. Open a terminal window.

c. Verify that Kali has an interface in the 10.6.6.0/24 network using the **ifconfig** command.

```
br-internal: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.6.6.1  netmask 255.255.255.0  broadcast 10.6.6.255
        inet6 fe80::42:bdff:fe50:acbe  prefixlen 64  scopeid 0x20<link>
        ether 02:42:bd:50:ac:be  txqueuelen 0  (Ethernet)
        RX packets 44  bytes 2536 (2.4 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 19  bytes 2732 (2.6 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

docker0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
```

d. Use the **nmap -V** command to verify that Nmap is installed and to display the Nmap version. The output will be similar to what is shown below.

```
┌──(kali㉿Kali)-[~]
└─$ nmap -V
```

Nmap version 7.93 (https://nmap.org)

Platform: x86_64-pc-linux-gnu

Compiled with: liblua-5.3.6 openssl-3.0.7 libssh2-1.10.0 libz-1.2.11 libpcre-8.39 nmap-libpcap-1.7.3 nmap-libdnet-1.12 ipv6

Compiled without:

Available nsock engines: epoll poll select

```
┌──(kali㉿Kali)-[~]
└─$ nmap -V
Nmap version 7.94 ( https://nmap.org )
Platform: x86_64-pc-linux-gnu
Compiled with: liblua-5.4.4 openssl-3.0.9 libssh2-1.10.0 libz-1.2.13 libpcre-
8.39 libpcap-1.10.4 nmap-libdnet-1.12 ipv6
Compiled without:
Available nsock engines: epoll poll select
```

Step 2: Investigate Nmap Options and Features

a. Using the command **nmap** without specifying any options or targets returns a list of commonly used Nmap options. To access the Nmap help system use the command **nmap -h**. The help output is divided into sections based on the type of detection that the option supports.

```
┌──(kali㉿Kali)-[~]
└─$ nmap -h
Nmap 7.94 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
  --excludefile <exclude_file>: Exclude list from file
HOST DISCOVERY:
  -sL: List Scan - simply list targets to scan
  -sn: Ping Scan - disable port scan
  -Pn: Treat all hosts as online -- skip host discovery
  -PS/PA/PU/PY[portlist]: TCP SYN/ACK, UDP or SCTP discovery to given ports
  -PE/PP/PM: ICMP echo, timestamp, and netmask request discovery probes
  -PO[protocol list]: IP Protocol Ping
  -n/-R: Never do DNS resolution/Always resolve [default: sometimes]
  --dns-servers <serv1[,serv2],...>: Specify custom DNS servers
  --system-dns: Use OS's DNS resolver
  --traceroute: Trace hop path to each host
SCAN TECHNIQUES:
  -sS/sT/sA/sW/sM: TCP SYN/Connect()/ACK/Window/Maimon scans
  -sU: UDP Scan
  -sN/sF/sX: TCP Null, FIN, and Xmas scans
```

b. The man page for Nmap provides additional information. To access the man page, enter the command **man nmap**. To exit the man pages, press **q** to quit and return to the terminal prompt.

```
NMAP(1)                    Nmap Reference Guide                    NMAP(1)

NAME
       nmap - Network exploration tool and security / port scanner

SYNOPSIS

       nmap [Scan Type ...] [Options] {target specification}

DESCRIPTION
       Nmap ("Network Mapper") is an open source tool for network
       exploration and security auditing. It was designed to rapidly scan
       large networks, although it works fine against single hosts. Nmap
       uses raw IP packets in novel ways to determine what hosts are
       available on the network, what services (application name and
       version) those hosts are offering, what operating systems (and OS
       versions) they are running, what type of packet filters/firewalls
       are in use, and dozens of other characteristics. While Nmap is
       commonly used for security audits, many systems and network
       administrators find it useful for routine tasks such as network
       inventory, managing service upgrade schedules, and monitoring host
       or service uptime.

       The output from Nmap is a list of scanned targets, with supplemental
       information on each depending on the options used. Key among that
       information is the "interesting ports table".  That table lists the
Manual page nmap(1) line 1 (press h for help or q to quit)
```

Use the man page for Nmap to complete the table.

Common NMAP Options

Option	Description
-A	**Aggressive scan options, this option enables additional advanced and aggressive options. Presently this enables OS detection (-O), version scanning (-sV), script scanning (-sC) and traceroute (--traceroute).**
-O	**Enable OS detection.**
-p <port ranges>	**Only scan specified ports.**
-sF	**FIN scan, sets just the TCP FIN bit.**

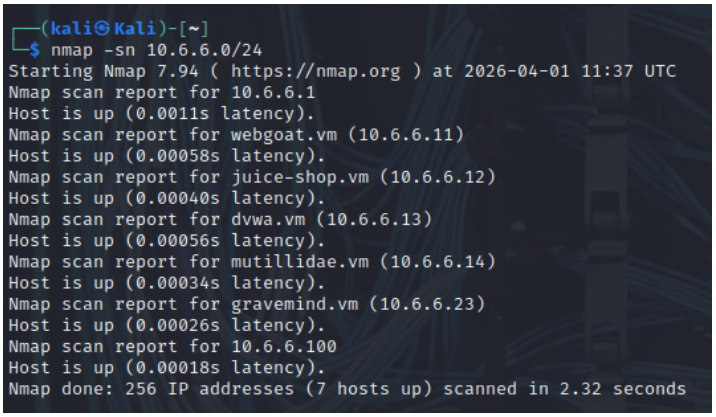
Option	Description
-sn	Null scan, does not set any bits.
-sS	TCP SYN scan, this technique is often referred to as half-open scanning or stealth scan, because you don't open a full TCP connection.
-sT	TCP connect scan is the default TCP scan type when SYN scan is not an option. This is the case when a user does not have raw packet privileges. Instead of writing raw packets as most other scan types do, Nmap asks the underlying operating system to establish a connection with the target machine and port by issuing the connect system call.
-sV	Enables version detection, as discussed above. Alternatively, you can use -A, which enables version detection among other things.
-T<0-5>	Set timing template (higher is faster).
-v	Increase verbosity level (use -vv or more for greater effect).
--open	Only show open (or possibly open) ports.

Part 2: Perform Basic Nmap Scans

Step 1: Initiate a basic Nmap scan of the target computer.

a. To quickly scan the DMZ for active hosts, you can perform a discovery scan. In a discovery scan, the scanning host sends an ICMP echo request (ping), a TCP SYN to port 443, a TCP ACK to port 80, and an ICMP timestamp request. A response to any of the requests indicates that the host is up and the IP protocol stack on the host is functioning. Enter the following command to scan the DMZ network:

```
┌──(kali㉿Kali)-[~]
└─$ nmap -sn 10.6.6.0/24
```



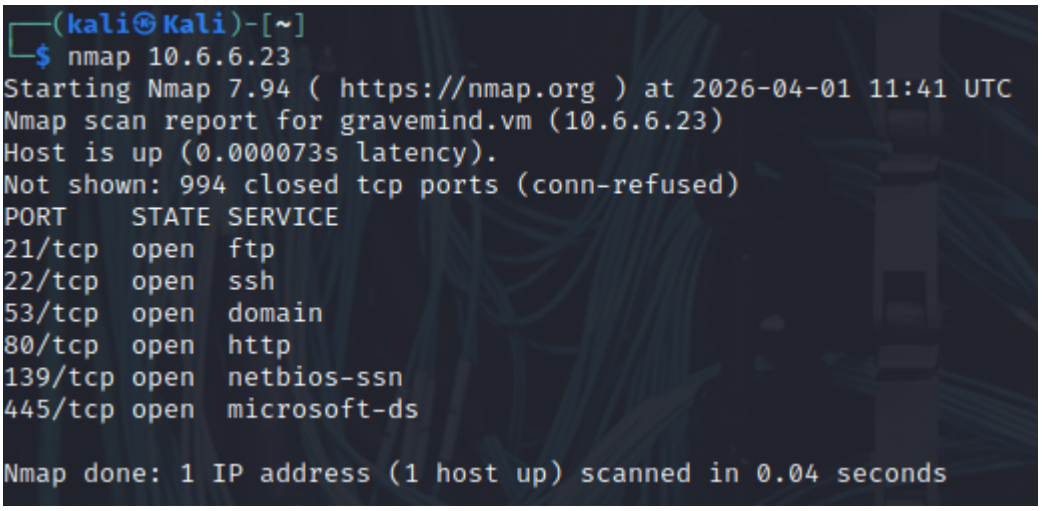
Question:

How many active hosts are located in the DMZ network?

In the scan it says there are 7 hosts up. But there are 5 hosts, which are webgoat.vm, juice-shop.vm, dvwa.vm, mutillidae.vm and gravemind.vm.

b. The host 10.6.6.23 was identified as suspicious in a Wireshark capture, and it is necessary to perform additional reconnaissance to discover more about the computer and its services. Use the **nmap** command to execute a default scan on the target host.

```
┌──(kali㉿Kali)-[~]
└─$ nmap 10.6.6.23
```



Question:

What ports are listed as open on the target host (10.6.6.23)?

Port 21 (FTP), port 22 (SSH), port 53 (DNS), port 80 (http), port 139 (SMB) and port 445 (also SMB) - new

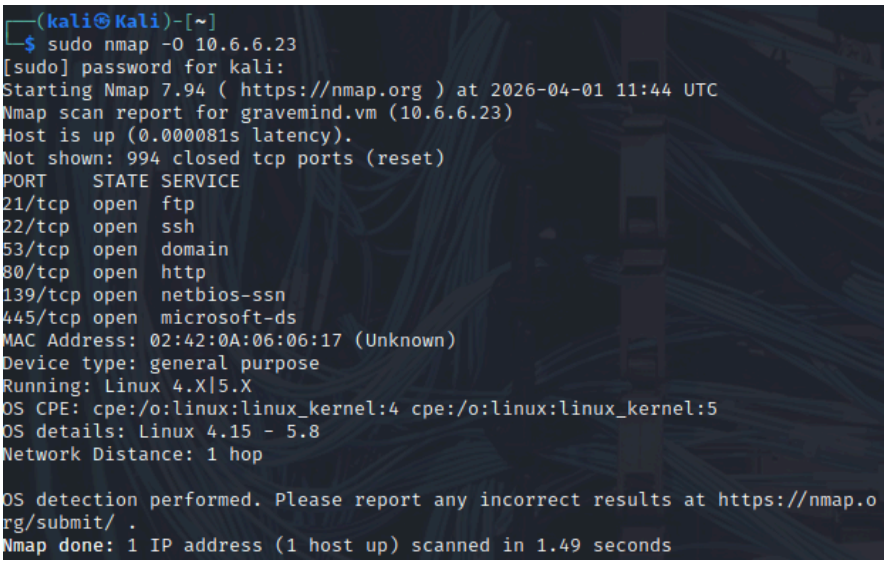
By default, Nmap performs a connect scan of 1000 most common TCP ports. This makes use of the operating system's networking software to establish a full TCP connection. This type of scan creates a lot of networking traffic and increases the probability of detection by intrusion detection services. You can also specify a TCP connect scan using the command option **nmap -sT**.

The output of the connect scan includes the status codes shown in the table:

Status	**Response Received**	**Interpretation**
Open	TCP SYN-ACK	There is a service listening on the identified port.
Closed	TCP RST	There is no service listening on the identified port.
Filtered	No response, or an ICMP destination unreachable message received.	The port is being filtered by a firewall.

c. The **-O** option can be used to further determine information about the operating system running on the target host. Some Nmap options require additional permissions and must be run as **root** or using the **sudo** command. To find operating system information on the target host, use the **nmap -O** command. Enter the password of **kali** when prompted.

┌──(kali㉿Kali)-[~]
└─$ **sudo nmap -O 10.6.6.23**



Question:

What operating system is the target host running?

Linux 4.15 - 5.8.

Step 2: Obtain additional information about the host and services.

a. To provide additional information about the target computer, it is possible to combine different options into a single command line. The previous command identified several potentially open ports on the 10.6.6.23 host. You can use **-v**, **-p**, and **-sV** to find additional information about the services running on the open ports. This command provides information about the FTP service running on port 21 on the target in verbose mode, with the timing set to fast (**-T4**):

┌──(kali㉿Kali)-[~]
└─$ **nmap -v -p21 -sV -T4 10.6.6.23**

Question:

What did you discover about the type and version of FTP server that is running on the host?

b. The **-A** option executes OS detection, version detection, script scanning, and traceroute. The **-A** scan can be very intrusive and therefore will be detected by many IDS systems, so ensure that you have permission before attempting this scan outside of the lab environment. To gather more information regarding the FTP service, enter the command **nmap -A -p21 10.6.6.23**.

The sample detailed output of this command is shown below:

┌──(kali㉿Kali)-[~]

└─$ **nmap -p21 -sV -A 10.6.6.23**

Starting Nmap 7.93 (https://nmap.org) at 2023-03-16 22:36 UTC

Nmap scan report for 10.6.6.23

Host is up (0.00044s latency).

PORT STATE SERVICE VERSION

21/tcp open ftp vsftpd 3.0.3

| ftp-syst:

| STAT:

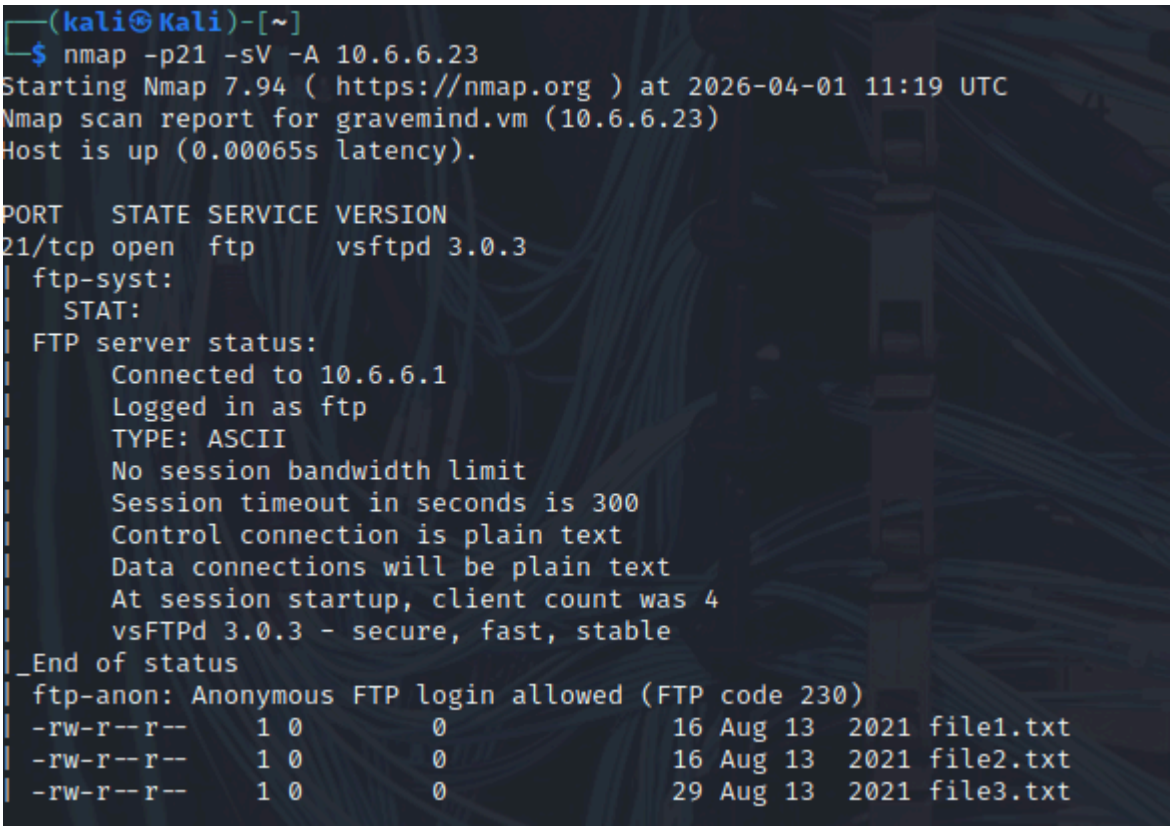
| FTP server status:

| Connected to 10.6.6.1

| Logged in as ftp

```
|      TYPE: ASCII
|      No session bandwidth limit
|      Session timeout in seconds is 300
|      Control connection is plain text
|      Data connections will be plain text
|      At session startup, client count was 3
|      vsFTPd 3.0.3 - secure, fast, stable
|_End of status
| ftp-anon: Anonymous FTP login allowed (FTP code 230)
| -rw-r--r--    1 0        0             16 Aug 13  2021 file1.txt
| -rw-r--r--    1 0        0             16 Aug 13  2021 file2.txt
| -rw-r--r--    1 0        0             29 Aug 13  2021 file3.txt
|_-rw-r--r--    1 0        0             26 Aug 13  2021 supersecretfile.txt
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 0.63 seconds
```



c. Examine the output of the **nmap -A** command. Notice that the output indicates that a connection was made between the Kali Linux system and the target FTP service.

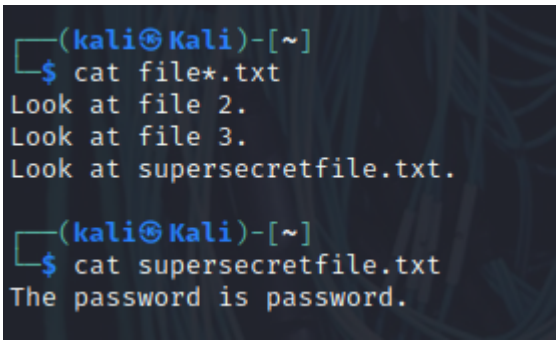
Questions:

How many files on the FTP server are accessible through this connection?

Four files, file1.txt, file2.txt, file3.txt and supersecretfile.txt.

What weakness in the FTP server configuration enabled the Kali Linux system to log into the FTP server?

FTP Anonymous login is allowed, in which an attacker or user can login with username "Anonymous" but without putting a password.



Files can be retrieved using the command "get <filename>".

Step 3:Investigate SMB Services with Scripts

The Server Message Block (SMB) protocol is a network file sharing protocol supported on Windows computers and by SAMBA on Linux. SMB enables applications to read and write files or request services over a network. Open public shares or shared devices such as print servers on a network, can be accessed through SMB.

a. The earlier scan of open ports on the target computer indicates that the SMB ports 139 and 445 are open. Find more information on these ports using the **-A** and **-p** command options. The **-A** option executes several functions including running the default scripts. Specify more than one port to scan by listing them separately with a comma between them.

┌──(kali㉿Kali)-[~]
└─$ **nmap -A -p139,445 10.6.6.23**
Starting Nmap 7.93 (https://nmap.org) at 2023-03-01 22:47 UTC
Nmap scan report for 10.6.6.23
Host is up (0.00014s latency).

PORT STATE SERVICE VERSION
139/tcp open netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open netbios-ssn Samba smbd 4.9.5-Debian (workgroup: WORKGROUP)
Service Info: Host: 868CF29B394C

Host script results:
| smb2-time:
| date: 2023-03-01T22:47:38
|_ start_date: N/A
| smb-security-mode:
| account_used: <blank>
| authentication_level: user
| challenge_response: supported
|_ message_signing: disabled (dangerous, but default)
| smb-os-discovery:
| OS: Windows 6.1 (Samba 4.9.5-Debian)
| Computer name: 868cf29b394c
| NetBIOS computer name: 868CF29B394C\x00
| Domain name: \x00
| FQDN: 868cf29b394c
|_ System time: 2023-03-01T22:47:35+00:00
| smb2-security-mode:

| 311:
|_ Message signing enabled but not required

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.05 seconds

```
┌──(kali㉿Kali)-[~]
└─$ nmap -A -p139,445 10.6.6.23
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-01 11:51 UTC
Nmap scan report for gravemind.vm (10.6.6.23)
Host is up (0.00046s latency).

PORT    STATE SERVICE     VERSION
139/tcp open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp open  netbios-ssn Samba smbd 4.9.5-Debian (workgroup: WORKGROUP)
Service Info: Host: GRAVEMIND

Host script results:
| smb-security-mode:
|   account_used: <blank>
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| smb-os-discovery:
|   OS: Windows 6.1 (Samba 4.9.5-Debian)
|   Computer name: gravemind
|   NetBIOS computer name: GRAVEMIND\x00
|   Domain name: \x00
|   FQDN: gravemind
|_  System time: 2026-04-01T11:52:01+00:00
| smb2-time:
```

b. Examine the information returned by the Nmap scan. From this information, it can be determined that the target computer is a member of the default workgroup, named WORKGROUP, and that SMB supported on this host through SAMBA on Linux.

Question:

What is the NetBIOS computer name assigned to the target host?

gravemind

c. Nmap contains the powerful Nmap Scripting Engine (NSE), which enables the programming of various Nmap options and conditional actions to be taken as a result of the responses. NSE has built-in scripts that enumerate users, groups, and network shares. One of the more commonly used scripts for SMB discovery is the **smb-enum-users.nse** script. Use the Nmap NSE script with the command:

┌──(kali㉿Kali)-[~]
└─$ **nmap --script smb-enum-users.nse -p139,445 10.6.6.23**

```
┌──(kali㉿Kali)-[~]
└─$ nmap --script smb-enum-users.nse -p139,445 10.6.6.23
Starting Nmap 7.94 ( https://nmap.org ) at 2026-04-01 11:55 UTC
Nmap scan report for gravemind.vm (10.6.6.23)
Host is up (0.00071s latency).

PORT    STATE SERVICE
139/tcp open  netbios-ssn
445/tcp open  microsoft-ds

Host script results:
| smb-enum-users:
|   GRAVEMIND\arbiter (RID: 1001)
|     Full name:
|     Description:
|     Flags:       Password not required, Account disabled, Normal user accou
nt
|   GRAVEMIND\masterchief (RID: 1000)
|     Full name:
|     Description:
|_    Flags:       Password not required, Account disabled, Normal user accou
nt

Nmap done: 1 IP address (1 host up) scanned in 0.16 seconds
```

Question:

Did the script uncover any SMB usernames on the target host? If so, how many?

Yes, two usernames. arbiter and masterchief.

d. A serious security concern is the existence of publicly shared directories (folders). You can enumerate the network shares using another NSE script, **smb-enum-shares.nse.** To discover shared directories on the target computer. Use the Nmap share enumeration script with the command:

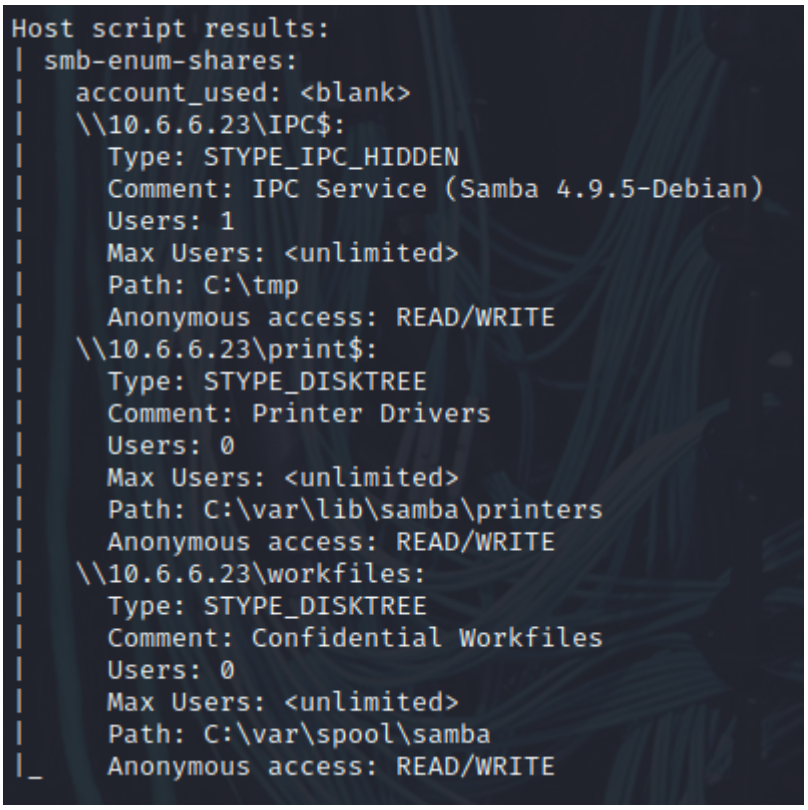
```
┌──(kali㉿Kali)-[~]
└─$ nmap --script smb-enum-shares.nse -p445 10.6.6.23
Starting Nmap 7.93 ( https://nmap.org ) at 2023-03-01 22:52 UTC
Nmap scan report for 10.6.6.23
Host is up (0.00016s latency).

PORT    STATE SERVICE
445/tcp open  microsoft-ds

Host script results:
| smb-enum-shares:
|   account_used: <blank>
|   \\10.6.6.23\IPC$:
|     Type: STYPE_IPC_HIDDEN
|     Comment: IPC Service (Samba 4.9.5-Debian)
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\tmp
|     Anonymous access: READ/WRITE
|   \\10.6.6.23\print$:
|     Type: STYPE_DISKTREE
|     Comment: Printer Drivers
|     Users: 0
|     Max Users: <unlimited>
|     Path: C:\var\lib\samba\printers
|     Anonymous access: READ/WRITE
|   \\10.6.6.23\workfiles:
|     Type: STYPE_DISKTREE
|     Comment: Confidential Workfiles
|     Users: 0
|     Max Users: <unlimited>
|     Path: C:\var\spool\samba
|_    Anonymous access: READ/WRITE

Nmap done: 1 IP address (1 host up) scanned in 12.92 seconds
```

e. Examine the output created by the **smb-enum-shares** script. In the output, share names that end with a "$" character represent hidden shares that include system and administrative shares.



```
Host script results:
| smb-enum-shares:
|   account_used: <blank>
|   \\10.6.6.23\IPC$:
|     Type: STYPE_IPC_HIDDEN
|     Comment: IPC Service (Samba 4.9.5-Debian)
|     Users: 1
|     Max Users: <unlimited>
|     Path: C:\tmp
|     Anonymous access: READ/WRITE
|   \\10.6.6.23\print$:
|     Type: STYPE_DISKTREE
|     Comment: Printer Drivers
|     Users: 0
|     Max Users: <unlimited>
|     Path: C:\var\lib\samba\printers
|     Anonymous access: READ/WRITE
|   \\10.6.6.23\workfiles:
|     Type: STYPE_DISKTREE
|     Comment: Confidential Workfiles
|     Users: 0
|     Max Users: <unlimited>
|     Path: C:\var\spool\samba
|_    Anonymous access: READ/WRITE
```

Questions:

How many hidden shares were discovered on the target host?

Three (3) which are IPC, printer and workfiles.

What serious security risk is uncovered in this script output?

All detected shares allow anonymous READ/WRITE access. This means an attacker doesn't need a password to connect using smbclient. Once connected, they can steal confidential files or upload a malicious script (like a reverse shell) directly into the share's path.

There are preprogrammed scripts that can provide additional SMB discovery capabilities if an authorized user account is available. Take time and investigate some of the NSE scripts that enumerate Windows and SAMBA systems.

Reflection Questions

1. Nmap is a powerful tool for network discovery. Think about the ways that Nmap can discover and enumerate computers that you used in this lab. How can Nmap be used by internal network technicians to inventory and secure local computers? How can these same tools be used by malicious actors to perform reconnaissance before an attack?

   Technicians are using Nmap to maintain the "Network Hygiene" of their organization, by scanning and enumerating for a leaked path, exposed files and what is connected to the server. Additionally, they can maintain the asset inventory by running sweep scans to find "Shadow IT" like unauthorized devices or forgotten servers that haven't been patched.

   Periodically scanning to ensure only "approved" services (like SSH or HTTPS) are running and that outdated, insecure protocols (like Telnet or SMBv1) have been disabled.

2. If you were tasked with creating a report on the status of the target host (10.6.6.23), what serious security risks would you include in your report?

   For an attacker, Nmap is the "Eyes" of the operation. It is used during the reconnaissance phase to build a map of the "Attack Surface." Using -sV, an attacker finds the exact version of a service (for example Samba 4.9.5). They can then search databases like CVE Details for known exploits that work specifically against that version.

   Attackers may use Nmap to test how "loud" they can be before an Intrusion Detection System (IDS) alerts the admins, adjusting their scan speed and timing to stay under the radar. Using the Nmap Scripting Engine (NSE) to automate the discovery of low-hanging fruit, such as default passwords, heartbleed vulnerabilities and unauthenticated file shares.

-THE END-